原版PT的MFVI推导

符号约定

d: `dim_z`

m: `dim_g`

n: `seq_len`

r: `ternary_rank`

势函数

$\phi_u(Z_i = a) = \exp(S_{w_i,a})$
$\phi_t(H_i = k, Z_i = a, Z_j = b) = \exp(T_{a,b})(H_i = j, \text{否则为}1)$
$\phi_b(Z_i = a, G_i = g) = \exp(B_{g,a})$

联合概率分布

给定序列w（坐标为$i \in 1 \sim n$）,给定各位置i的Z_i, G_i, H_i，此取值概率为：$P(Z, H, G|w) = \text{Normalization}\{\Pi_{i=1}^n \phi_u(Z_i) * \Pi_{i=1}^n \phi_b(Z_i, G_i) * \Pi_{c=1}^h \Pi_{i=1}^n \Pi_{j=1}^n \phi_t(H_i^{(c)}, Z_i, Z_j)\}$

MFVI

变分自由能函数：$F(Q) = E_Q[-\log P(Z, H, G|w)] - H(Q)$
$= -\sum_{i=1}^n (\sum_{a=1}^d Q_i(Z_i = a) S_{w_i,a})$
$-\sum_{c=1}^C \sum_{i=1}^n (\sum_{j \neq i} \sum_{a=1}^d \sum_{b=1}^d Q_{ic}(H_i = j) Q_i(Z_i = a) Q_j(Z_j = b) T_{a,b}^{(c)})$
$-\sum_{i=1}^n (\sum_{a=1}^d \sum_{g=1}^m Q_i(Z_i = a) Q_i^G(G_i = g) B_{g,a})$
$-H(Q)$

MFVI: 求$Q_i(a)$分布使得$F(Q)$取得最小值，且$\sum_a Q_i(a) = 1$。以此构建Lagrange函数，使其对$Q_i(a)$偏导为0，解出一次迭代后的$Q_i(a)$。

Lagrange:$L = F(Q) + \lambda(\sum_z Q_i(a) - 1)$
偏导为0$\Rightarrow \nabla E - \ln Q_i(a) - 1 + \lambda = 0$ (其中∇E为$E_Q[-\log P(Z, H, G|w)]$对$Q_i(a)$的偏导，$\ln Q_i(a) + 1$为$H(Q)$对$Q_i(a)$的偏导)

$\nabla E_{Q_{ic}(j)}$的求解与$Q_{ic}(j)$的更新：

$\frac{\partial E}{\partial Q_{ic}(j)} = -\sum_a \sum_b Q_i(a)Q_j(b)T_{a,b}(c)$
$= -\sum_a \sum_b Q_i(a)Q_j(b)(\sum_l U_{a,l}V_{b,l})$
$= -\sum_l (\sum_a Q_i(a)U_{a,l})(\sum_b Q_j(b)V_{b,l})$
令$\sum_a Q_i(a)U_{a,l} = q_{i,l}, \sum_b Q_j(b)V_{b,l} = k_{j,l}$
则$\nabla E_{Q_{ic}(j)} = -\sum_{l=1}^{r} q_{i,l}k_{j,l} = -(\vec{q_i}\vec{k_j^T})$

由$Q \propto \exp(-\nabla E)$得：$Q_{ic}(j) \propto \exp(\vec{q_i}\vec{k_j^T})$

$\nabla E_{Q_i(a)}$的求解与$Q_i(a)$的更新：

对F,只考虑含$Q_i(a)$的项：
$F_i(Q) = -Q_i(a)S_{w_i,a}$
$-\sum_c \sum_{j\neq i} Q_{ic}(j) \sum_b Q_i(a)Q_j(b)T_{ab}^{(c)}$
$-\sum_c \sum_{k\neq i} Q_{kc}(i) \sum_b Q_k(b)Q_i(a)T_{ba}^{(c)}$
$-\sum_g Q_i(a)Q_i^G(g)B_{g,a}$

$\nabla_{unary} = -S_{w_i,a}$
$\nabla_{child} = -\sum_c \sum_{j\neq i} Q_{ic}(j)(\sum_b Q_j(b)T_{a,b}^{(c)})$
$\nabla_{head} = -\sum_c \sum_{k\neq i} Q_{kc}(i)(\sum_b Q_k(b)T_{b,a}^{(c)})$
$\nabla_{binary} = -\sum_g Q_i^G(g)G_{g,a}$
$\therefore \nabla E = \nabla_{unary} + \nabla_{child} + \nabla_{head} + \nabla_{binary}$
代入Lagrange函数，令其偏导为0$\Rightarrow \ln Q_i(a) = -\nabla E + 1 - \lambda$
$\Rightarrow Q_i(a) \propto \exp(-\nabla E)$
$\Rightarrow Q_i(a) = \mathrm{Softmax}(-\nabla E)$

即：$\mathrm{Logits}_i(a) = -\nabla E$
$= S_{w_i,a} + \sum_c \sum_{j\neq i} Q_{ic}(j)(\sum_b Q_j(b)T_{a,b}^{(c)}) + \sum_c \sum_{k\neq i} Q_{kc}(i)(\sum_b Q_k(b)T_{b,a}^{(c)}) +$
$\sum_g Q_i^G(g)B_{g,a}$

考虑T矩阵分解：

$\sum_{j\neq i} Q_{ic}(j)(\sum_b Q_j(b)T_{a,b}^{(c)}) = \sum_j \mathrm{Attn}_{ij}(Q_jT) = \sum_j \mathrm{Attn}_{ij}(Q_jV)U^T$
$\sum_{k\neq i} Q_{kc}(i)(\sum_b Q_k(b)T_{b,a}^{(c)}) = \sum_k \mathrm{Attn}_{ki}(Q_kT) = \sum_k \mathrm{Attn}_{ki}(Q_kU)V^T$

相应代码： `qz_new = Softmax(Input_S + (Attn_Child + Attn_Head) + FFN_Output)`

muP版 PT的MFVI推导

势函数 (muP Scaled)

$\tau = \frac{d}{r}$，为Q_i分布的温度参数。Q_{ic}分布的温度参数仍为1。后续进行推导解释。
$\phi_u(Z_i = a) = \exp(\tau S_{w_i,a})$
$\phi_t(H_i = k, Z_i = a, Z_j = b) = \exp(\tau d T_{a,b})(H_i = j, \text{否则为}1)$
$\phi_b(Z_i = a, G_i = g) = \exp(\tau m B_{g,a})$

联合概率分布

给定序列w（坐标为$i \in 1 \sim n$）,给定各位置i的Z_i, G_i, H_i，此取值概率为：$P(Z, H, G|w) = \text{Normalization}\{\Pi_{i=1}^n \phi_u(Z_i) * \Pi_{i=1}^n \phi_b(Z_i, G_i) * \Pi_{c=1}^h \Pi_{i=1}^n \Pi_{j=1}^n \phi_t(H_i^{(c)}, Z_i, Z_j)\}$

MFVI

变分自由能函数：$F(Q) = E_Q[-\log P(Z, H, G|w)] - \tau H(Q)$(添加了系数$\tau$，后续进行推导)
$= -\sum_{i=1}^n (\sum_{a=1}^d Q_i(Z_i = a)\tau S_{w_i,a})$
$-\sum_{c=1}^C \sum_{i=1}^n (\sum_{j \neq i} \sum_{a=1}^d \sum_{b=1}^d Q_{ic}(H_i = j) Q_i(Z_i = a) Q_j(Z_j = b)(\tau d T_{a,b}^{(c)}))$
$-\sum_{i=1}^n (\sum_{a=1}^d \sum_{g=1}^m Q_i(Z_i = a) Q_i^G(G_i = g)(\tau m B_{g,a}))$
$-\tau H(Q)$

MFVI: 求$Q_i(a)$分布使得$F(Q)$取得最小值，且$\sum_a Q_i(a) = 1$。以此构建Lagrange函数，使其对$Q_i(a)$偏导为0，解出一次迭代后的$Q_i(a)$。

Lagrange:$L = F(Q) + \lambda(\sum_z Q_i(a) - 1)$
偏导为0$\Rightarrow \nabla E - \tau(\ln Q_i(a) + 1) + \lambda = 0$ (其中∇E为$E_Q[-\log P(Z, H, G|w)]$对$Q_i(a)$的偏导，$\ln Q_i(a) + 1$为$H(Q)$对$Q_i(a)$的偏导)

$\nabla E_{Q_{ic}(j)}$的求解与$Q_{ic}(j)$的更新：

$\frac{\partial E}{\partial Q_{ic}(j)} = -\sum_a \sum_b Q_i(a) Q_j(b)(\tau d T_{a,b}(c))$
$= -\tau d \sum_a \sum_b Q_i(a) Q_j(b)(\sum_l U_{a,l} V_{b,l})$
$= -\tau d \sum_l (\sum_a Q_i(a) U_{a,l})(\sum_b Q_j(b) V_{b,l})$
令$\sum_a Q_i(a) U_{a,l} = q_{i,l}, \sum_b Q_j(b) V_{b,l} = k_{j,l}$
则$\nabla E_{Q_{ic}(j)} = -\tau d \sum_{l=1}^r q_{i,l} k_{j,l} = -(\frac{d^2}{r} \vec{q_i} \vec{k_j^T})$

由$Q \propto \exp(-\nabla E)$得：$Q_{ic}(j) \propto \exp(\frac{d^2}{r} \vec{q_i} \vec{k_j^T})$

$\nabla E_{Q_i(a)}$的求解与$Q_i(a)$的更新：

对F,只考虑含$Q_i(a)$的项：
$F_i(Q) = -Q_i(a)\tau S_{w_i,a}$
$-\sum_c \sum_{j\neq i} Q_{ic}(j) \sum_b Q_i(a)Q_j(b)(\tau dT_{ab}^{(c)})$
$-\sum_c \sum_{k\neq i} Q_{kc}(i) \sum_b Q_k(b)Q_i(a)(\tau dT_{ba}^{(c)})$
$-\sum_g Q_i(a)Q_i^G(g)(\tau mB_{g,a})$

$\nabla_{unary} = -\tau S_{w_i,a}$
$\nabla_{child} = -\sum_c \sum_{j\neq i} Q_{ic}(j)(\sum_b Q_j(b)\tau dT_{a,b}^{(c)})$
$\nabla_{head} = -\sum_c \sum_{k\neq i} Q_{kc}(i)(\sum_b Q_k(b)\tau dT_{b,a}^{(c)})$
$\nabla_{binary} = -\sum_g Q_i^G(g)(\tau mB_{g,a})$
$\therefore \nabla E = \nabla_{unary} + \nabla_{child} + \nabla_{head} + \nabla_{binary}$
代入Lagrange函数，令其偏导为0$\Rightarrow \ln Q_i(a) = -\frac{1}{\tau}\nabla E + 1 - \frac{1}{\tau}\lambda$
$\Rightarrow Q_i(a) \propto \exp(-\frac{1}{\tau}\nabla E)$
$\Rightarrow Q_i(a) = \mathrm{Softmax}(-\frac{1}{\tau}\nabla E)$

即：$\mathrm{Logits}_i(a) = -\frac{1}{\tau}\nabla E$
$= \frac{1}{\tau}[\tau S_{w_i,a} + \tau d\sum_c \sum_{j\neq i} Q_{ic}(j)(\sum_b Q_j(b)T_{a,b}^{(c)}) + \tau d\sum_c \sum_{k\neq i} Q_{kc}(i)(\sum_b Q_k(b)T_{b,a}^{(c)}) + \tau m\sum_g Q_i^G(g)B_{g,a}]$
$= S_{w_i,a} + d\sum_c \sum_{j\neq i} Q_{ic}(j)(\sum_b Q_j(b)T_{a,b}^{(c)}) + d\sum_c \sum_{k\neq i} Q_{kc}(i)(\sum_b Q_k(b)T_{b,a}^{(c)}) + m\sum_g Q_i^G(g)B_{g,a}$

考虑T矩阵分解：

$\sum_{j\neq i} Q_{ic}(j)(\sum_b Q_j(b)T_{a,b}^{(c)}) = \sum_j \mathrm{Attn}_{ij}(Q_jT) = \sum_j \mathrm{Attn}_{ij}(Q_jV)U^T$
$\sum_{k\neq i} Q_{kc}(i)(\sum_b Q_k(b)T_{b,a}^{(c)}) = \sum_k \mathrm{Attn}_{ki}(Q_kT) = \sum_k \mathrm{Attn}_{ki}(Q_kU)V^T$

代码实现的契合性检验

有关特殊缩放的代码：

```
class SquaredSoftmax(nn.Module):
    def __init__(self, dim=-1, eps=1e-6):
        super().__init__()
        self.dim = dim
        self.eps = eps

    def forward(self, hidden_states: torch.Tensor):
        input_dtype = hidden_states.dtype
        hidden_states = hidden_states.to(torch.float32)
        hidden_states = hidden_states.pow(2)
        # Energy Recovery: scale by dim
        hidden_states = F.normalize(hidden_states, p=1, dim=self.dim, eps=self.eps) * hidden_sta
        return hidden_states.to(input_dtype)


class AbsNormalization(nn.Module):
    def __init__(self, dim=-1, eps=1e-6):
        super().__init__()
        self.dim = dim
        self.eps = eps

    def forward(self, hidden_states: torch.Tensor):
        input_dtype = hidden_states.dtype
        hidden_states = hidden_states.to(torch.float32)
        hidden_states = F.relu(hidden_states)
        # Energy Recovery: scale by dim
        hidden_states = F.normalize(hidden_states, p=1, dim=self.dim, eps=self.eps) * hidden_sta
        return hidden_states.to(input_dtype)
```

即： `potential_function_z` 为 `square` , `potential_function_g` 为 `abs` ，归一化时增加了 `* hidden_states.shape[self.dim]`

原始代码中，涉及到归一化操作的变量为： `qz` ：相当于$Q_i(a)$； `qg` ：相当于$Q_i^G(g)$。原始代码中，经过归一化相当于： `qz` 与 `qg` 显式代表了概率分布。

修改后的代码中： `qz` 和 `qg` 分别乘上了d和m，即不再显示代表概率分布，而是代表概率分布乘宽度。

结合MFVI推导：

$\nabla E_{Q_{ic}(j)}$的求解与$Q_{ic}(j)$的更新：

$$\frac{\partial E}{\partial Q_{ic}(j)} = -\sum_a \sum_b Q_i(a)Q_j(b)(\frac{d^2}{r}T_{a,b}(c))$$
$$= -\frac{1}{r}\sum_a \sum_b (dQ_i(a))(dQ_j(b))(T_{a,b}(c))$$

$= -\frac{1}{r}\sum_l(\sum_a dQ_i(a)U_{a,l})(\sum_b dQ_j(b)V_{b,l})$
令 $\sum_a dQ_i(a)U_{a,l} = q_{i,l}, \sum_b dQ_j(b)V_{b,l} = k_{j,l}$
则 $\nabla E_{Q_{ic}(j)} = -\frac{1}{r}\sum_{l=1}^{r} q_{i,l}k_{j,l} = -(\frac{1}{r}\vec{q}_i\vec{k_j^T})$

此时 `qz` 即代表$dQ_i(a)$整体。原始代码中的 `Message_F` 即为$-\vec{q}_i\vec{k_j^T}$。为保持与MFVI推断的一致性，需将 `Message_F` 额外除以r（ternary_rank）。此时代码中 `Message_F` 即代表$-(\frac{1}{r}\vec{q}_i\vec{k_j^T})$，归一化后得$Q_ic(j)$，对应的变量为 `qh` 。

其余与原始MFVI推导一致。

$\nabla E_{Q_i(a)}$的求解与$Q_i(a)$的更新：

$\mathrm{Logits}_i(a) = -\frac{1}{\tau}\nabla E$
$= \frac{1}{\tau}[\tau S_{w_i,a} + \tau d\sum_c\sum_{j\neq i} Q_{ic}(j)(\sum_b Q_j(b)T_{a,b}^{(c)}) + \tau d\sum_c\sum_{k\neq i} Q_{kc}(i)(\sum_b Q_k(b)T_{b,a}^{(c)}) + \tau m\sum_g Q_i^G(g)B_{g,a}]$
$= \frac{1}{\tau}[\tau S_{w_i,a} + \tau\sum_c\sum_{j\neq i} Q_{ic}(j)(\sum_b dQ_j(b)T_{a,b}^{(c)}) + \tau\sum_c\sum_{k\neq i} Q_{kc}(i)(\sum_b dQ_k(b)T_{b,a}^{(c)}) + \tau\sum_g mQ_i^G(g)B_{g,a}]$
$= S_{w_i,a} + \sum_c\sum_{j\neq i} Q_{ic}(j)(\sum_b dQ_j(b)T_{a,b}^{(c)}) + \sum_c\sum_{k\neq i} Q_{kc}(i)(\sum_b dQ_k(b)T_{b,a}^{(c)}) + \sum_g mQ_i^G(g)B_{g,a}$

其中dQ_j,$dQ_k(b)$,$mQ_i^G(g)$仍对应代码中的 `qz` `qg` ，这两者仍然不再显示表示概率分布，但上式最终计算结果为概率分布。

Conclusion: 经过调整后，仍然能够符合MFVI的推导步骤。区别在于缩放因子被合入相关变量中。

代码总共进行了两处与原始PT不同的特殊缩放：
1.Normalization函数中全部乘 `self.dim[-1]` ，对应了 `qz` `qg` 意义的改变
2. `Message_F` 额外除以r，对应了凑齐$\nabla E_{Q_{ic}(j)}$的表达式。

两个温度参数τ与1合理性的推导